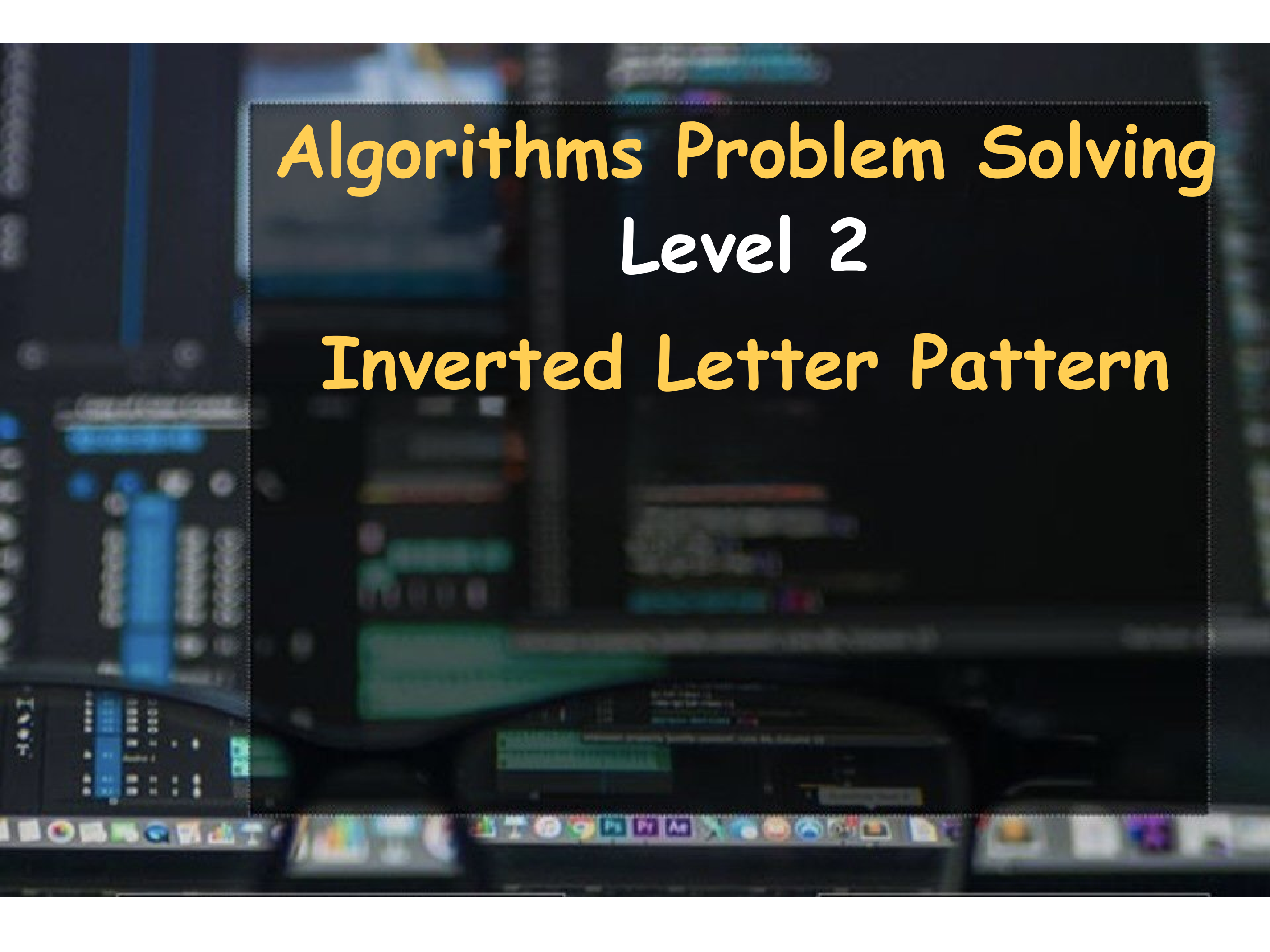
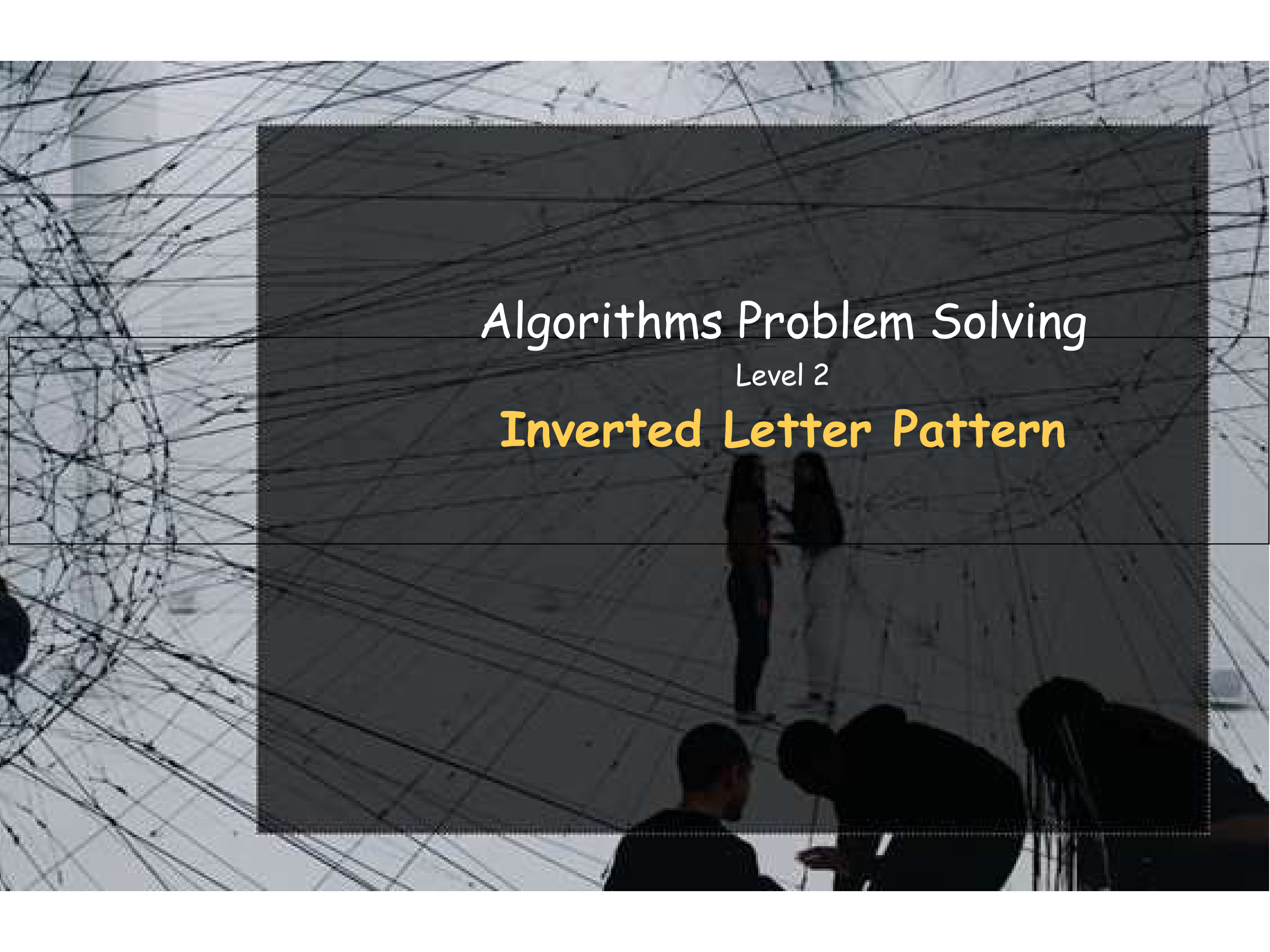




Algorithms Problem Solving

Level 2

Inverted Letter Pattern



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Inverted Letter Pattern

🔍 Definition

- ◆ An inverted letter pattern is a pattern where letters are printed from a higher letter down to `A`, while the number of repetitions decreases in each row.
- ◆ For example, input `3` produces `CCC`, `BB`, `A`.

Explanation

- ◆ ``ReadNumber()`` reads a non-negative number from the user.
- ◆ ``PrintInvertedLetterPattern()`` uses two nested ``for`` loops.
- ◆ ``65`` represents the character code of uppercase ``A``.
- ◆ ``65 + Number - 1`` calculates the starting character.
- ◆ For ``Number = 3``:
- ◆ ``65 + 3 - 1 = 67`` $\rightarrow$ ``C``.
- ◆ The outer loop starts from this character and decreases until ``65`` (``A``).
- ◆ ``char(i)`` converts the numeric character code into a letter.
- ◆ The inner loop controls how many times the current letter is printed.
- ◆ The number of repetitions decreases from ``Number`` down to ``1``.

Steps in main()

1. Read a non-negative number.
2. Calculate the starting character using `'65 + Number - 1'`.
3. Start the outer loop from the starting character down to `'A'`.
4. Convert `'i'` to a character using `'char(i)'`.
5. Use the inner loop to print the current letter the required number of times.
6. Move to the next line.
7. Decrease the letter and the number of repetitions.

💡 Example

◆ Input:

`3`

◆ Starting character:

`65 + 3 - 1 = 67 → C`

◆ First row: `C` printed 3 times.

◆ Second row: `B` printed 2 times.

◆ Third row: `A` printed 1 time.

➡ Result

CCC

BB

A

Note

- ◆ The ASCII value of uppercase `A` is `65`, `B` is `66`, `C` is `67`, and so on.
- ◆ `char(i)` converts the numeric value back into its corresponding character.
- ◆ The expression `Number - ((65 + Number - 1) - i)` calculates how many times the current letter should be printed.

⊘ Warning

- ◆ The program allows `0` because the validation condition is `N < 0`.
- ◆ If the input is larger than the range of uppercase letters, the pattern can go beyond `Z` and produce other characters.
- ◆ The code relies on the character encoding values for uppercase English letters.



Comparison

- ▨ #12_Inverted_Number_Pattern
 - ✓ Prints numbers in descending order.
 - ✓ Example: `3 → 333, 22, 1`.

- ▤ #14_Inverted_Letter_Pattern
 - ✓ Prints letters in descending order.
 - ✓ Example: `3 → CCC, BB, A`.

✦✦ Summary

- ✓ Use ``65`` as the character code for ``A``.
- ✓ Calculate the starting letter with ``65 + Number - 1``.
- ✓ Use the outer loop to move backward from the starting letter to ``A``.
- ✓ Use the inner loop to control repetitions.
- ✓ ``char(i)`` converts the numeric character code into a letter.
- ✓ Input ``5`` produces ``EEEEEE``, ``DDDD``, ``CCC``, ``BB``, ``A``.

Thank you 😊